

Asymptotic Inversion of the Subfactorial

Bell-Number Tails, Inverse-Gamma Anchoring, and a Calculus for Exponentially Separated Transseries

Vladimir Reshetnikov

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Abstract

Let $D_n = !n$ denote the number of derangements of n objects. The integer sequence itself has no unique continuous inverse, so the first task is to specify an interpolation. We use the real median of the two lateral incomplete-gamma continuations,

$$\mathfrak{D}(x) = \frac{\Gamma(x+1)}{e} + \cos(\pi x)J(x), \quad J(x) = \int_0^1 e^{-t}(1-t)^x dt, \quad x > -1.$$

It satisfies $\mathfrak{D}(n) = D_n$, is strictly increasing for $x \geq 3$, and therefore has a principal inverse $\mathfrak{X}(y)$ for $y \geq 2$. The coefficient J has the convergent inverse-factorial representation

$$J(x) = \sum_{r \geq 0} \frac{(-1)^r}{(x+1)_{r+1}}$$

and the Bell-number expansion

$$J(x) \sim \sum_{k \geq 1} (-1)^{k-1} \frac{B_k}{x^k}, \quad 0 < (-1)^M \left(J(x) - \sum_{k=1}^M (-1)^{k-1} \frac{B_k}{x^k} \right) \leq \frac{B_{M+1}}{x^{M+1}}.$$

This extends the known integer formula to every real $x > 0$, with its sharp sign and constant.

Write $G(x) = \Gamma(x+1)/e$ and let $g = G^{-1}$ be its increasing large branch. With

$$A = \log \frac{ey}{\sqrt{2\pi}}, \quad w = W_0(A/e), \quad X = \frac{A}{w} = e^{w+1}, \quad L = \log X,$$

we derive the outer inverse-gamma expansion

$$g(y) = X - \frac{1}{2} + \sum_{m \geq 1} \frac{a_m(L)}{X^{2m-1}},$$

including an explicit triangular recurrence and the first five rational coefficient blocks. The subfactorial correction is beyond every algebraic order in X^{-1} . Conjugating by G and applying the Lagrange–Bürmann formula gives the all-orders exact-coefficient transseries

$$\mathfrak{X}(y) = g(y) + \sum_{n \geq 1} \frac{(-1)^n}{n!} \left(\frac{d}{dy} \right)^{n-1} [g'(y)q(g(y))^n], \quad q(x) = \cos(\pi x)J(x).$$

Its first sector is

$$-\frac{\cos(\pi g(y))}{y \psi(g(y)+1)} \left(\frac{1}{g(y)} - \frac{2}{g(y)^2} + \frac{5}{g(y)^3} - \dots \right),$$

and the n -th sector has size $\mathcal{O}(y^{-n}g(y)^{-n}/\log g(y))$ for fixed n . We also prove a universal operator formula for inverting $G(x) + \sum_j \sigma_j G(x)^{\alpha_j} h_j(x)$, with sector grade $\sum_j m_j(1 - \alpha_j)$. The article treats interpolation ambiguity, nested late-term effects of the Bell expansion, convergence and remainder conditions, symbolic generation, numerical validation, residual certificates, and recovery of the integer generalized inverse.

Keywords. Subfactorial; derangement; Bell numbers; inverse gamma function; Lambert W ; logarithmic transseries; Lagrange–Bürmann inversion; asymptotic reversion; exponentially small corrections; incomplete gamma function.

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1 The inverse problem and the answer in three layers

The subfactorial or derangement number is

$$D_n = !n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}, \quad n \in \mathbb{N}_0. \quad (1)$$

For $n \geq 1$, the familiar estimate $D_n \approx n!/e$ is extraordinarily accurate: indeed, D_n is the nearest integer to $n!/e$. Hassani's integral identity refines this by isolating the signed difference and leads to the Bell-number expansion

$$D_n = \frac{n!}{e} + \sum_{k=1}^M (-1)^{n+k-1} \frac{B_k}{n^k} + \mathcal{O}(n^{-M-1}), \quad (2)$$

with implied constant at most B_{M+1} ; see [1]. The purpose of this article is to invert the full object, not merely its factorial leading term.

There are three asymptotic layers.

1. The dominant equation $\Gamma(x+1)/e = y$ is inverted by a Lambert- W -anchored logarithmic expansion. This is the *zero sector*.
2. The Bell tail contributes a correction of order $y^{-1}x^{-1}(\log x)^{-1}$, oscillating through $\cos(\pi x)$. It is smaller than every power of the zero-sector variable X^{-1} .
3. Repeated reversion creates sectors of order $y^{-2}, y^{-3}, \dots$. Each sector is a finite Fourier polynomial whose coefficients have power-logarithmic expansions generated from the Bell numbers and polygamma functions.

These layers must not be collapsed into a single ordinary Poincaré series.

An integer sequence does not determine a unique continuous inverse

Knowing D_n at integer arguments does not determine a function between the integers. If $R(x)$ is any sufficiently small smooth function, then

$$\tilde{\mathfrak{D}}(x) = \mathfrak{D}(x) + \sin(\pi x)R(x)$$

has the same values at every integer but a different beyond-all-orders inverse transseries. Consequently, the zero-sector inverse-gamma expansion is robust, whereas the oscillatory sectors depend on the chosen interpolation. We make the interpolation explicit before speaking of an inverse.

Main result

Define

$$\mathfrak{D}(x) = G(x) + q(x), \quad G(x) = \frac{\Gamma(x+1)}{e}, \quad q(x) = \cos(\pi x)J(x), \quad (3)$$

where

$$J(x) = \int_0^1 e^{-t}(1-t)^x dt. \quad (4)$$

Then $\mathfrak{D}(n) = D_n$ for every $n \geq 0$, and $\mathfrak{D}$ is strictly increasing on $[3, \infty)$. Let

$$\mathfrak{X} = (\mathfrak{D}|_{[3, \infty)})^{-1} : [2, \infty) \longrightarrow [3, \infty). \quad (5)$$

If $g = G^{-1}$ is the increasing large branch, then

$$\mathfrak{X}(y) = g(y) + \sum_{n \geq 1} \mathcal{T}_n(y), \quad \mathcal{T}_n(y) = \frac{(-1)^n}{n!} \left(\frac{d}{dy} \right)^{n-1} [g'(y)q(g(y))^n]. \quad (6)$$

For every fixed N , this is an asymptotic expansion through sector N ; under the analytic hypotheses stated in [Section 7](#), it is the Taylor series at perturbation parameter zero and converges at parameter one for all sufficiently large y .

The rest of the paper derives (6), expands all its ingredients, and generalizes it to a broad class of rapidly increasing dominant functions.

2 A canonical real interpolation of the subfactorial

2.1 From Hassani's integral to a median continuation

Hassani's identity can be written

$$D_n = \frac{n!}{e} + \frac{(-1)^n}{e} L_n(e), \quad L_n(e) = \int_1^e (\log t)^n dt. \quad (7)$$

The substitutions $t = e^u$ and then $u = 1 - s$ give

$$\frac{1}{e} L_n(e) = \frac{1}{e} \int_0^1 u^n e^u du = \int_0^1 e^{-s} (1-s)^n ds = J(n), \quad (8)$$

which is real and positive for $x > -1$. Replacing $(-1)^n$ by $\cos(\pi x)$ gives (3).

There is also a useful complex interpretation. Continue the upper incomplete gamma integral to the point -1 along the upper or lower bank of the negative real axis. For $\Re z > -1$, splitting the contour at the origin gives the two lateral continuations

$$\mathfrak{D}_\pm(z) = \frac{1}{e} \Gamma(z+1, -1)_\pm = \frac{\Gamma(z+1)}{e} + e^{\pm i\pi z} J(z). \quad (9)$$

Their real median on the real axis is

$$\mathfrak{D}(x) = \frac{1}{2} (\mathfrak{D}_+(x) + \mathfrak{D}_-(x)) = G(x) + \cos(\pi x) J(x). \quad (10)$$

At integer n , both lateral values coincide because $e^{\pm i\pi n} = (-1)^n$.

Proposition 2.1 (Interpolation property). *For every integer $n \geq 0$,*

$$\mathfrak{D}(n) = D_n. \quad (11)$$

Moreover, $\mathfrak{D}$ is real analytic on $(-1, \infty)$.

Proof. Equation (11) follows immediately from (7)–(8). Real analyticity of J on $x > -1$ follows by differentiating under the integral sign on compact subintervals; the gamma and cosine factors are analytic there as well. $\square$

2.2 Monotonicity and the principal inverse

The following elementary bounds are enough to isolate an unambiguous large branch.

Lemma 2.2 (Bounds for the Bell tail and its derivatives). *For $x > 0$ and every integer $r \geq 0$,*

$$\left| J^{(r)}(x) \right| \leq \frac{r!}{(x+1)^{r+1}}. \quad (12)$$

In particular,

$$\left| q'(x) \right| \leq \frac{\pi}{x+1} + \frac{1}{(x+1)^2}. \quad (13)$$

Proof. Set $s = 1 - e^{-u}$ in (4). Then

$$J(x) = \int_0^\infty e^{-xu} K(u) \, du, \quad K(u) = e^{-u+e^{-u}-1}. \quad (14)$$

Since $K(u) \leq e^{-u}$ for $u \geq 0$, differentiation under the integral gives

$$\left| J^{(r)}(x) \right| \leq \int_0^\infty u^r e^{-(x+1)u} \, du = \frac{r!}{(x+1)^{r+1}}.$$

Finally, $q' = -\pi \sin(\pi x)J + \cos(\pi x)J'$, which yields (13). $\square$

Theorem 2.3 (Existence of the principal inverse). *The function $\mathfrak{D}$ is strictly increasing on $[3, \infty)$, maps that interval onto $[2, \infty)$, and therefore has the inverse $\mathfrak{X}$ defined in (5).*

Proof. The dominant derivative is

$$G'(x) = G(x)\psi(x+1), \quad (15)$$

where ψ is the digamma function. For $x \geq 3$, both $\Gamma(x+1)$ and $\psi(x+1)$ are increasing, and hence

$$G'(x) \geq \frac{6}{e}\psi(4) = \frac{6}{e} \left(\frac{11}{6} - \gamma \right) > 2.7.$$

By (13),

$$|q'(x)| \leq \frac{\pi}{4} + \frac{1}{16} < 0.85.$$

Thus $\mathfrak{D}'(x) = G'(x) + q'(x) > 0$ on $[3, \infty)$. Since $\mathfrak{D}(3) = D_3 = 2$ and $\mathfrak{D}(x) \rightarrow \infty$, the image is $[2, \infty)$. $\square$

Corollary 2.4 (Integer generalized inverse). *For $y \geq 2$, define*

$$N(y) = \max\{n \in \mathbb{N} : n \geq 3, D_n \leq y\}. \quad (16)$$

Then

$$N(y) = \lfloor \mathfrak{X}(y) \rfloor. \quad (17)$$

In particular, $\mathfrak{X}(D_n) = n$ for every $n \geq 3$.

Proof. Strict monotonicity gives $D_n = \mathfrak{D}(n) \leq y < D_{n+1} = \mathfrak{D}(n+1)$ exactly when $n \leq \mathfrak{X}(y) < n+1$. $\square$

3 The Bell-number coefficient: exact forms, asymptotics, and late terms

The coefficient $J(x)$ is the bridge between the discrete derangement identity and the continuous inverse transseries. It is advantageous to retain several equivalent representations because they serve different purposes: the integral gives positivity and derivative bounds, the inverse-factorial series is convergent and numerically stable, and the Bell series exposes the power asymptotics.

3.1 A convergent inverse-factorial series

Theorem 3.1 (Exact representations of J). *For every real $x > -1$,*

$$J(x) = \int_0^1 e^{-t} (1-t)^x dt, \quad (18)$$

$$= \frac{1}{e} \int_0^1 e^u u^x du, \quad (19)$$

$$= \sum_{r=0}^{\infty} \frac{(-1)^r}{(x+1)_{r+1}}, \quad (20)$$

$$= \int_0^{\infty} e^{-xu} e^{-u+e^{-u}-1} du. \quad (21)$$

The series (20) converges absolutely. For $x > 0$, its alternating truncation error is bounded by the first omitted term:

$$\left| J(x) - \sum_{r=0}^{R-1} \frac{(-1)^r}{(x+1)_{r+1}} \right| \leq \frac{1}{(x+1)_{R+1}}. \quad (22)$$

Proof. Equivalence of (18) and (19) follows from $u = 1 - t$. Expanding e^{-t} in (18) and integrating beta integrals yields

$$\int_0^1 \frac{(-t)^r}{r!} (1-t)^x dt = \frac{(-1)^r}{r!} \frac{\Gamma(r+1)\Gamma(x+1)}{\Gamma(x+r+2)} = \frac{(-1)^r}{(x+1)_{r+1}}.$$

Absolute convergence follows from comparison with a factorial tail. For $x > 0$, the magnitudes decrease strictly, so the alternating-series bound gives (22). Finally, the substitution $t = 1 - e^{-u}$ gives (21). $\square$

The recurrence

$$t_0 = \frac{1}{x+1}, \quad t_{r+1} = -\frac{t_r}{x+r+2}, \quad J(x) = \sum_{r \geq 0} t_r, \quad (23)$$

computes J without large intermediate quantities. It is preferable to the divergent Bell expansion when many digits are needed.

3.2 Why Bell numbers appear

Set

$$K(u) = e^{-u+e^{-u}-1}. \quad (24)$$

The exponential mixture

$$K(u) = \frac{1}{e} \sum_{r=0}^{\infty} \frac{e^{-(r+1)u}}{r!} \quad (25)$$

shows that K is completely monotone on $[0, \infty)$.

Lemma 3.2 (Bell derivatives). *For every integer $m \geq 0$,*

$$K^{(m)}(0) = (-1)^m B_{m+1}. \quad (26)$$

Proof. Differentiate (25) termwise:

$$K^{(m)}(0) = \frac{(-1)^m}{e} \sum_{r=0}^{\infty} \frac{(r+1)^m}{r!} = \frac{(-1)^m}{e} \sum_{k=1}^{\infty} \frac{k^{m+1}}{k!}.$$

Dobinski's formula $B_j = e^{-1} \sum_{k \geq 0} k^j / k!$ gives the result. $\square$

Theorem 3.3 (Bell expansion with a signed remainder). *For every integer $M \geq 1$ and every real $x > 0$,*

$$J(x) = \sum_{k=1}^M (-1)^{k-1} \frac{B_k}{x^k} + R_M(x), \quad (27)$$

where

$$0 < (-1)^M R_M(x) \leq \frac{B_{M+1}}{x^{M+1}}. \quad (28)$$

Consequently,

$$J(x) \sim \frac{1}{x} - \frac{2}{x^2} + \frac{5}{x^3} - \frac{15}{x^4} + \frac{52}{x^5} - \frac{203}{x^6} + \frac{877}{x^7} - \dots. \quad (29)$$

Proof. Taylor's formula at the origin gives

$$K(u) = \sum_{m=0}^{M-1} \frac{K^{(m)}(0)}{m!} u^m + \rho_M(u).$$

Complete monotonicity implies

$$0 < (-1)^M \rho_M(u) \leq \frac{(-1)^M K^{(M)}(0)}{M!} u^M = \frac{B_{M+1}}{M!} u^M.$$

Insert this formula into the Laplace representation (21). Because $\int_0^\infty e^{-xu} u^m \, du = m!/x^{m+1}$, [Theorem 3.2](#) gives (27) and (28). $\square$

Corollary 3.4 (Hassani's expansion, with the same bound). *For integers $n \geq 1$,*

$$D_n = \frac{n!}{e} + \sum_{k=1}^M (-1)^{n+k-1} \frac{B_k}{n^k} + (-1)^n R_M(n), \quad (30)$$

where $|R_M(n)| \leq B_{M+1}/n^{M+1}$.

Proof. Use $D_n = n!/e + (-1)^n J(n)$ and [Theorem 3.3](#). $\square$

The derivative expansions follow by termwise differentiation at any fixed order.

Corollary 3.5 (Derivatives of the Bell tail). *For each fixed $r \geq 0$,*

$$J^{(r)}(x) \sim (-1)^r \sum_{k \geq 1} (-1)^{k-1} \frac{(k)_r B_k}{x^{k+r}}, \quad x \rightarrow +\infty, \quad (31)$$

where $(k)_r = k(k+1) \cdots (k+r-1)$ is the rising factorial.

3.3 Late terms and the nested transseries warning

The Bell series is asymptotic, not convergent. This matters when it is placed inside the exponentially small inverse sectors. The standard Bell-number estimate may be written in the form

$$B_m \sim \frac{N^m e^{N-m-1}}{\sqrt{1 + \log N}}, \quad N \log N = m, \quad (32)$$

with higher corrections available; see [\[4, 7\]](#). The ratio heuristic places the least term of B_m/x^m near

$$m_\star = x \log x, \quad N = x. \quad (33)$$

At this index, (32) gives

$$\frac{B_{m_\star}}{x^{m_\star}} \asymp \frac{e^{-x \log x + x - 1}}{\sqrt{1 + \log x}} \asymp \frac{\sqrt{x/\log x}}{G(x)}. \quad (34)$$

More precisely, the quotient of the two displayed leading scales tends to $\sqrt{2\pi}/e^2$.

The Bell coefficient expansion invades the next exponential sector

At $x = g(y)$, one has $G(x) = y$. If the first inverse sector $-q/(y\psi)$ is evaluated by optimally truncating the Bell series for J , then the induced least-term uncertainty is of order

$$\frac{1}{y \log x} \text{rac} \sqrt{x/\log xy} = \mathcal{O}\left(\frac{\sqrt{x}}{y^2(\log x)^{3/2}}\right). \quad (35)$$

It therefore lies in the y^{-2} exponential grade and can be larger than the explicit reversion-generated y^{-2} block. A fully expanded transseries is nested: late terms of one coefficient sector mix with later exponential sectors. To preserve clean grading, retain J exactly through (18) or the convergent series (20), and expand it in Bell numbers only to the accuracy actually required.

This phenomenon is not a defect. It is the expected interaction between a power asymptotic expansion and a second, exponentially smaller scale. The general calculus in Section 6 is therefore formulated first with exact coefficient functions and only then with their internal transseries.

4 The zero sector: inversion of $\Gamma(x+1)/e$

Let

$$G(x) = \frac{\Gamma(x+1)}{e}, \quad g(y) = G^{-1}(y) \quad (36)$$

where g denotes the increasing large branch. We first construct a useful Lambert- W -anchored expansion for g .

4.1 Centered Stirling expansion

Set

$$u = x + \frac{1}{2}. \quad (37)$$

The centered Stirling expansion is

$$\log \Gamma(x+1) \sim u(\log u - 1) + \frac{1}{2} \log(2\pi) + \sum_{m \geq 1} \frac{\kappa_m}{u^{2m-1}}, \quad (38)$$

where

$$\kappa_m = -\frac{(1 - 2^{1-2m})B_{2m}^{\text{Ber}}}{2m(2m-1)}. \quad (39)$$

Here B_{2m}^{Ber} denotes a Bernoulli number; the superscript is included only to prevent confusion with the Bell numbers B_k . The first terms are

$$\kappa_1 = -\frac{1}{24}, \quad \kappa_2 = \frac{7}{2880}, \quad \kappa_3 = -\frac{31}{40320}, \quad \kappa_4 = \frac{127}{215040}. \quad (40)$$

This centered form, obtained from the standard shifted gamma expansion, has only odd inverse powers and is especially convenient for reversion; see [3, 9].

The equation $G(x) = y$ becomes

$$u(\log u - 1) + \sum_{m \geq 1} \frac{\kappa_m}{u^{2m-1}} = A, \quad A = \log \frac{ey}{\sqrt{2\pi}}. \quad (41)$$

Ignore the inverse powers temporarily and solve

$$X(\log X - 1) = A. \quad (42)$$

With the principal Lambert function,

$$w = W_0(A/e), \quad X = \frac{A}{w} = e^{w+1}, \quad L = \log X = w + 1. \quad (43)$$

Indeed, $X(L-1) = Xw = A$.

4.2 Triangular reversion around the Lambert anchor

Write

$$u = X(1 + \eta), \quad t = X^{-1}, \quad \eta(t) = \sum_{m \geq 1} a_m(L) t^{2m}. \quad (44)$$

Substitution in (41) yields the single formal identity

$$0 = t^{-1} \{ \eta(L-1) + (1 + \eta) \log(1 + \eta) \} + \sum_{r \geq 1} \kappa_r t^{2r-1} (1 + \eta)^{-(2r-1)}. \quad (45)$$

At order t^{2m-1} , the unknown a_m occurs only in the linear term La_m . This gives a triangular algorithm.

Algorithm 4.1 (Outer inverse-gamma coefficients). Suppose $a_1, \dots, a_{m-1}$ are known. Put

$$\eta_{<m}(t) = \sum_{j=1}^{m-1} a_j(L) t^{2j}$$

and let $\rho_m(L)$ be the coefficient of t^{2m-1} in the right-hand side of (45) after replacing η by $\eta_{<m} + a_m t^{2m}$, with the term La_m omitted. Then

$$a_m(L) = -\frac{\rho_m(L)}{L}. \quad (46)$$

Equivalently, expand (45) through order t^{2m-1} and solve the resulting linear equation for a_m .

Theorem 4.2 (Lambert-anchored inverse-gamma expansion). As $y \rightarrow +\infty$, with A, w, X, L defined by (43),

$$g(y) \sim X - \frac{1}{2} + \sum_{m \geq 1} \frac{a_m(L)}{X^{2m-1}}. \quad (47)$$

The first five coefficient blocks are

$$a_1(L) = \frac{1}{24L}, \quad (48)$$

$$a_2(L) = -\frac{14L^2 + 10L + 5}{5760L^3}, \quad (49)$$

$$a_3(L) = \frac{2232L^4 + 1176L^3 + 714L^2 + 350L + 105}{2903040L^5}, \quad (50)$$

$$a_4(L) = -\frac{1}{1393459200L^7} (822960L^6 + 292536L^5 + 136956L^4 + 68040L^3 + 32970L^2 + 12250L + 2625), \quad (51)$$

$$a_5(L) = \frac{1}{367873228800L^9} (309052800L^8 + 77920128L^7 + 26024592L^6 + 10019592L^5 + 4516028L^4 + 2023560L^3 + 812350L^2 + 242550L + 40425). \quad (52)$$

After truncation through a_M/X^{2M-1} , the asymptotic error is

$$\mathcal{O}\left(\frac{1}{LX^{2M+1}}\right). \quad (53)$$

Moreover, $a_m(L)$ is a rational polynomial in L^{-1} , beginning at order L^{-1} and ending at order $L^{-(2m-1)}$.

Proof. Substitution of (44) into the centered Stirling equation gives (45). Because

$$(1 + \eta) \log(1 + \eta) = \eta + \frac{1}{2}\eta^2 - \frac{1}{6}\eta^3 + \cdots,$$

the coefficient multiplying the new a_m is $(L - 1) + 1 = L$; all other contributions depend only on previous coefficients. This proves triangularity and (46). Straight expansion gives (48)–(52). The next omitted term in the centered Stirling series is $\mathcal{O}(X^{-(2M+1)})$, while the derivative of $u(\log u - 1)$ at $u = X$ is $\log X = L$. Division by this derivative gives (53). The stated rational structure follows inductively from (45). $\square$

4.3 Conversion to a pure logarithmic expansion

The Lambert function itself has the standard expansion

$$\begin{aligned} W_0(z) \sim L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2(-2 + L_2)}{2L_1^2} \\ + \frac{L_2(6 - 9L_2 + 2L_2^2)}{6L_1^3} + \frac{L_2(-12 + 36L_2 - 22L_2^2 + 3L_2^3)}{12L_1^4} + \cdots, \end{aligned} \quad (54)$$

where $L_1 = \log z$ and $L_2 = \log L_1$; see [5, 2]. Substituting $z = A/e$ into (54), then using $X = A/W_0(A/e)$, turns (47) into a conventional logarithmic transseries in $A, \log A, \log \log A$. The Lambert-anchored form is usually superior: it resums the entire leading logarithmic inversion exactly and leaves only odd powers of X^{-1} .

Remark 4.3 (What the zero sector can and cannot see). Since

$$y = G(g(y)) \asymp \exp(g(y) \log g(y) - g(y)),$$

one has $y^{-1} = o(X^{-N})$ for every fixed N . Thus no number of terms in (47) can reveal the Bell correction. The latter is a separate exponential sector, not a further coefficient in the same X^{-1} series.

5 The full inverse subfactorial transseries

We now restore the additive correction $q(x) = \cos(\pi x)J(x)$. The cleanest derivation is to conjugate the subfactorial interpolation by the dominant map G .

5.1 Conjugation to a near-identity problem

Let

$$z = G(x), \quad x = g(z), \quad Q(z) = q(g(z)). \quad (55)$$

Then

$$\mathfrak{D}(g(z)) = z + Q(z). \quad (56)$$

If h denotes the inverse of $z \mapsto z + Q(z)$, then

$$\mathfrak{X}(y) = g(h(y)). \quad (57)$$

The generalized Lagrange–Bürmann formula applied to the composition $g \circ h$ gives the answer in one line.

Theorem 5.1 (Exact-coefficient inverse transseries). *As a formal series in a perturbation parameter ε , the inverse of*

$$\mathfrak{D}_\varepsilon(x) = G(x) + \varepsilon q(x) \quad (58)$$

has the expansion

$$\mathfrak{X}_\varepsilon(y) = g(y) + \sum_{n \geq 1} \varepsilon^n \mathcal{T}_n(y), \quad (59)$$

where

$$\mathcal{T}_n(y) = \frac{(-1)^n}{n!} \left(\frac{d}{dy} \right)^{n-1} [g'(y)q(g(y))^n]. \quad (60)$$

Equivalently, if $A_1(x) = G'(x)$, then

$$\mathcal{T}_n(y) = \frac{(-1)^n}{n!} \left[\left(A_1(x)^{-1} \frac{d}{dx} \right)^{n-1} \left(\frac{q(x)^n}{A_1(x)} \right) \right]_{x=g(y)}. \quad (61)$$

Setting $\varepsilon = 1$ gives the inverse subfactorial transseries (6).

Proof. In the conjugated variable, the perturbed equation is

$$y = z + \varepsilon Q(z), \quad Q(z) = q(g(z)).$$

The Lagrange–Bürmann composition formula gives

$$g(h_\varepsilon(y)) = g(y) + \sum_{n \geq 1} \frac{(-\varepsilon)^n}{n!} \left(\frac{d}{dy} \right)^{n-1} [g'(y)Q(y)^n],$$

which is (60). Since $g'(y) = 1/G'(g(y))$ and $d/dy = G'(x)^{-1} d/dx$ at $x = g(y)$, one obtains (61). $\square$

Formula (61) is an algorithm, not merely an existence statement. It differentiates only known functions and automatically produces the correct nonlinear combinations of q , gamma derivatives, and polygamma functions.

5.2 The first three exponential sectors

For compactness, write

$$A_1 = G', \quad A_2 = G'', \quad A_3 = G'''. \quad (62)$$

Direct expansion of (61) gives

$$\mathcal{T}_1 = -\frac{q}{A_1}, \quad (63)$$

$$\mathcal{T}_2 = \frac{qq'}{A_1^2} - \frac{q^2 A_2}{2A_1^3}, \quad (64)$$

$$\mathcal{T}_3 = -\frac{q^2 q''}{2A_1^3} - \frac{q(q')^2}{A_1^3} + \frac{q^3 A_3}{6A_1^4} + \frac{3q^2 q' A_2}{2A_1^4} - \frac{q^3 A_2^2}{2A_1^5}, \quad (65)$$

where every quantity on the right is evaluated at $x = g(y)$.

Introduce the polygamma abbreviations

$$p = \psi(x+1), \quad p_1 = \psi^{(1)}(x+1), \quad p_2 = \psi^{(2)}(x+1). \quad (66)$$

At $x = g(y)$, $G(x) = y$, and therefore

$$G' = yp, \quad (67)$$

$$G'' = y(p^2 + p_1), \quad (68)$$

$$G''' = y(p^3 + 3pp_1 + p_2). \quad (69)$$

Consequently,

$$\boxed{\mathcal{T}_1(y) = -\frac{q(x)}{yp}, \quad x = g(y).} \quad (70)$$

The next two blocks are

$$\mathcal{T}_2(y) = \frac{1}{y^2} \left[\frac{qq'}{p^2} - \frac{q^2(p^2 + p_1)}{2p^3} \right]_{x=g(y)}, \quad (71)$$

$$\mathcal{T}_3(y) = \frac{1}{y^3} \left[-\frac{q^2q''}{2p^3} - \frac{q(q')^2}{p^3} + \frac{q^3(p^3 + 3pp_1 + p_2)}{6p^4} + \frac{3q^2q'(p^2 + p_1)}{2p^4} - \frac{q^3(p^2 + p_1)^2}{2p^5} \right]_{x=g(y)}. \quad (72)$$

For actual computation one may use

$$q = cJ, \quad (73)$$

$$q' = -\pi sJ + cJ', \quad (74)$$

$$q'' = -\pi^2 cJ - 2\pi sJ' + cJ'', \quad (75)$$

where $c = \cos(\pi x)$, $s = \sin(\pi x)$.

5.3 The explicit first Bell–Fourier sector

The digamma expansion is

$$\psi(x+1) \sim \log x + \frac{1}{2x} - \frac{1}{12x^2} + \frac{1}{120x^4} - \frac{1}{252x^6} + \cdots. \quad (76)$$

Combining it with (29) gives a power-logarithmic coefficient expansion for the first exponential sector. Let $\ell = \log x$. Then

$$\begin{aligned} \frac{J(x)}{\psi(x+1)} &\sim \frac{1}{x\ell} - \frac{4\ell+1}{2x^2\ell^2} + \frac{60\ell^2+13\ell+3}{12x^3\ell^3} \\ &\quad - \frac{360\ell^3+64\ell^2+14\ell+3}{24x^4\ell^4} \\ &\quad + \frac{37440\ell^4+5694\ell^3+1025\ell^2+225\ell+45}{720x^5\ell^5} + \mathcal{O}\left(\frac{1}{x^6\ell}\right). \end{aligned} \quad (77)$$

Thus, with $x = g(y)$,

$$\mathfrak{X}(y) = g(y) - \frac{\cos(\pi g(y))}{y} \left[\frac{1}{g \log g} - \frac{4 \log g + 1}{2g^2(\log g)^2} + \frac{60(\log g)^2 + 13 \log g + 3}{12g^3(\log g)^3} - \cdots \right] + \mathcal{T}_2(y) + \mathcal{T}_3(y) + \cdots. \quad (78)$$

Here and only here, inside the square brackets, g abbreviates $g(y)$. Replacing $g(y)$ by the Lambert-anchored series (47) produces a transseries entirely in $W_0(A/e)$, inverse powers of X , powers of y^{-1} , and the Fourier phase $e^{\pm i\pi g(y)}$.

The leading form of the first three sectors is especially simple:

$$\mathcal{T}_n(y) \sim -\frac{q(g(y))^n}{n y^n \psi(g(y)+1)} \sim -\frac{\cos^n(\pi g(y))}{n y^n g(y)^n \log g(y)}, \quad (79)$$

for each fixed n , away from accidental zeros of the displayed leading coefficient. The second equivalence uses $J(x) \sim x^{-1}$.

5.4 Sector size and Fourier support

Theorem 5.2 (Subfactorial sector structure). *For each fixed $n \geq 1$,*

$$\mathcal{T}_n(y) = \mathcal{O}_n\left(\frac{1}{y^n g(y)^n \log g(y)}\right), \quad y \rightarrow +\infty. \quad (80)$$

After expanding J , the polygamma functions, and the zero sector, the coefficient of y^{-n} is a finite Fourier polynomial in $e^{i\pi g(y)}$ with frequencies

$$-n, -n+2, \dots, n-2, n. \quad (81)$$

Each Fourier coefficient is a power-logarithmic series in $g(y)^{-1}$ whose coefficients lie in the ring generated over $\mathbb{Q}$ by Bell numbers, Bernoulli numbers, and powers of π .

Proof. By Theorem 2.2, every fixed derivative of q is $\mathcal{O}_n(x^{-1})$. Moreover, $G' = G\psi$ and $\psi(x+1) \sim \log x$. Formula (61) therefore contributes one factor $G^{-1} = y^{-1}$ per application and starts with $q^n = \mathcal{O}(x^{-n})$, yielding (80). Writing

$$q(x) = \frac{1}{2}J(x) (e^{i\pi x} + e^{-i\pi x})$$

shows that q^n has precisely the frequencies in (81). Differentiation changes coefficients but not frequencies. Finally, the Bell expansion of J , the Bernoulli expansion of the polygamma functions, and differentiation generate the stated coefficient ring. $\square$

Remark 5.3 (Lateral rather than median interpolation). For $\mathfrak{D}_\pm = G + e^{\pm i\pi x}J$, the n -th sector carries a single frequency $\pm n\pi$. The real median combines the two lateral modes, and multinomial products produce the finite frequency fan (81).

6 A general inversion calculus for rapidly increasing transseries

The preceding calculation is an instance of a universal pattern. A rapidly increasing dominant map is inverted first; additive or relatively smaller transseries are then reverted by a near-identity Lagrange expansion. The logarithmic derivative of the dominant map controls every coefficient block.

6.1 The one-perturbation operator formula

Let G be a strictly increasing function on a large real interval and set

$$\lambda(x) = \frac{G'(x)}{G(x)} = \frac{d}{dx} \log G(x). \quad (82)$$

The elementary identity

$$\frac{1}{G\lambda} \frac{d}{dx} (G^{-k} f) = G^{-(k+1)} \left(\lambda^{-1} \frac{d}{dx} - k \right) f \quad (83)$$

is the source of the block structure. Define

$$\mathcal{L}_k = \lambda^{-1} \frac{d}{dx} - k. \quad (84)$$

Theorem 6.1 (Universal additive-block formula). *Let*

$$F(x) = G(x) + H(x), \quad (85)$$

and let $g = G^{-1}$. Formally, and asymptotically under the hypotheses of [Section 7](#),

$$F^{-1}(y) = g(y) + \sum_{n \geq 1} \frac{U_n(g(y))}{y^n}, \quad (86)$$

where

$$U_n(x) = \frac{(-1)^n}{n!} \mathcal{L}_{n-1} \mathcal{L}_{n-2} \cdots \mathcal{L}_1 \left(\frac{H(x)^n}{\lambda(x)} \right). \quad (87)$$

For $n = 1$, the operator product is empty. The first blocks are

$$U_1 = -\frac{H}{\lambda}, \quad (88)$$

$$U_2 = \frac{1}{2} (\lambda^{-1} D_x - 1) \left(\frac{H^2}{\lambda} \right), \quad (89)$$

$$U_3 = -\frac{1}{6} (\lambda^{-1} D_x - 2) (\lambda^{-1} D_x - 1) \left(\frac{H^3}{\lambda} \right). \quad (90)$$

Proof. Apply (61) with $q = H$ and $A_1 = G' = G\lambda$. The expression inside the iterated derivative is

$$\frac{H^n}{G\lambda} = G^{-1} \frac{H^n}{\lambda}.$$

Repeated use of (83), with $k = 1, 2, \dots, n-1$, produces

$$G^{-n} \mathcal{L}_{n-1} \cdots \mathcal{L}_1 (H^n / \lambda).$$

At $x = g(y)$, $G(x) = y$, which proves (86)–(87). $\square$

For the subfactorial, $H = q$ and

$$\lambda(x) = \psi(x+1). \quad (91)$$

Thus (87) is exactly the compact form of the full subfactorial sector calculus.

6.2 Several relative scales and a sector semigroup

The same factorization handles perturbations that carry powers of the dominant function. Consider

$$F(x) = G(x) + \sum_{j=1}^s \sigma_j G(x)^{\alpha_j} h_j(x), \quad \alpha_j < 1. \quad (92)$$

Let $\mathbf{m} = (m_1, \dots, m_s) \in \mathbb{N}_0^s \setminus \{0\}$, and define

$$n = |\mathbf{m}| = \sum_j m_j, \quad a = \mathbf{m} \cdot \boldsymbol{\alpha} = \sum_j m_j \alpha_j, \quad h_{\mathbf{m}} = \prod_j h_j^{m_j}, \quad \mathbf{m}! = \prod_j m_j!. \quad (93)$$

Theorem 6.2 (Multiscale inversion and sector grading). *The formal inverse of (92) has the expansion*

$$F^{-1}(y) = g(y) + \sum_{\mathbf{m} \neq 0} \boldsymbol{\sigma}^{\mathbf{m}} y^{-\nu(\mathbf{m})} U_{\mathbf{m}}(g(y)), \quad (94)$$

with sector grade

$$\nu(\mathbf{m}) = n - a = \sum_{j=1}^s m_j (1 - \alpha_j) > 0. \quad (95)$$

To compute the coefficient, set

$$V_0(x) = \frac{h_{\mathbf{m}}(x)}{\lambda(x)} \quad (96)$$

and recursively define

$$V_{r+1}(x) = (\lambda(x)^{-1} D_x - (1 - a + r)) V_r(x), \quad 0 \leq r \leq n - 2. \quad (97)$$

Then

$$U_{\mathbf{m}}(x) = \frac{(-1)^n}{\mathbf{m}!} V_{n-1}(x). \quad (98)$$

For $n = 1$, $V_{n-1} = V_0$.

Proof. In the n -th Lagrange block, the multinomial expansion of the perturbation gives

$$\left(\sum_j \sigma_j G^{\alpha_j} h_j \right)^n = \sum_{|\mathbf{m}|=n} \frac{n!}{\mathbf{m}!} \sigma^{\mathbf{m}} G^a h_{\mathbf{m}}.$$

After division by $G' = G\lambda$, the $\mathbf{m}$ -term is $G^{-(1-a)} h_{\mathbf{m}}/\lambda$. Apply (83) successively with $k = 1 - a, 2 - a, \dots, n - 1 - a$. The final power is $G^{-(n-a)}$, and the factor $n!$ cancels the one in the Lagrange coefficient, leaving (97)–(98). $\square$

How to read the grading rule

Every perturbing scale $G^{\alpha_j} h_j$ carries a positive defect $1 - \alpha_j$ relative to G . Multiplication adds defects, so the inverse sectors are indexed by the additive semigroup generated by $1 - \alpha_1, \dots, 1 - \alpha_s$. If two multiindices have the same defect, their coefficient blocks collide and must be added. Logarithms, algebraic powers, and Fourier phases remain in the coefficient algebra and do not alter the exponential grade.

6.3 Differential coefficient algebras and Fourier closure

A convenient formal setting is a differential algebra $\mathcal{A}$ of “slow” coefficients closed under multiplication, D_x , and multiplication by λ^{-1} . The block operators $\mathcal{L}_k$ then preserve $\mathcal{A}$. Typical choices include algebras generated by

$$x^{-1}, \quad (\log x)^{-1}, \quad \log \log x, \quad e^{i\omega_1 x}, \dots, e^{i\omega_r x}, \quad (99)$$

with formal power-logarithmic expansions in their amplitudes. Products add frequencies, so finite Fourier support remains finite in every fixed sector.

For the median subfactorial interpolation, take two coefficient modes

$$h_+(x) = \frac{1}{2} e^{i\pi x} J(x), \quad h_-(x) = \frac{1}{2} e^{-i\pi x} J(x), \quad \alpha_+ = \alpha_- = 0. \quad (100)$$

A multiindex (r, s) has grade $r + s$ and frequency $(r - s)\pi$, which immediately reproduces (81).

6.4 Nested coefficient transseries

Suppose a slow coefficient itself has a transseries

$$h_j(x) \sim \sum_{\beta \in S_j} \zeta(x)^\beta h_{j,\beta}(x), \quad \zeta(x) \rightarrow 0. \quad (101)$$

Substitution into (98) and collection of monomials enlarges the sector semigroup by the grades carried by ζ . Algebraically, the new support is the Minkowski sum of the outer inversion semigroup and the

internal supports of the coefficients. The Bell late-term effect in (35) is precisely such a collision: the hidden exponential scale in the asymptotic expansion of J enters at the same y^{-2} grade as the second Lagrange block.

The practical conclusion is important. A formally complete result should be stated first in exact-coefficient form, such as (60); only then should individual coefficient functions be asymptotically expanded, with a truncation rule that respects all active scales.

7 Validity, convergence, and remainder control

The Lagrange formulas have two complementary interpretations. They are identities in a formal perturbation parameter, and they are analytic Taylor series whenever the conjugated near-identity map is sufficiently small. The latter condition is automatically satisfied for the subfactorial at large positive arguments.

7.1 A local analytic criterion

Let

$$y = z + \varepsilon Q(z) \quad (102)$$

with Q analytic near a positive point y . The inverse value is the fixed point of

$$z = y - \varepsilon Q(z). \quad (103)$$

Proposition 7.1 (Convergence in the perturbation parameter). *Suppose that for some disc $B(y, R)$, constants $M, \kappa > 0$, and $\delta > 0$,*

$$\sup_{B(y, R)} |Q| \leq M, \quad \sup_{B(y, R)} |Q'| \leq \kappa, \quad (1 + \delta)M < R, \quad (1 + \delta)\kappa < 1. \quad (104)$$

Then for every $|\varepsilon| \leq 1 + \delta$, equation (103) has a unique solution in $B(y, R)$. This solution is analytic in ε , and its Taylor series at $\varepsilon = 0$ is the Lagrange–Bürmann series. In particular, the series converges at $\varepsilon = 1$.

Proof. The right-hand side of (103) maps the disc into itself and has Lipschitz constant at most $(1 + \delta)\kappa < 1$. The contraction mapping theorem gives the unique fixed point. Analytic dependence on ε follows from the analytic implicit function theorem because $1 + \varepsilon Q'(z) \neq 0$. The Taylor coefficients are those of the Lagrange–Bürmann formula; see, for example, [6]. $\square$

Proposition 7.2 (Application to the subfactorial). *For the canonical interpolation $\mathfrak{D} = G + q$, the series (59) converges at $\varepsilon = 1$ for every sufficiently large positive y . Moreover, for each fixed N , with $x = g(y)$,*

$$\mathfrak{X}(y) = g(y) + \sum_{n=1}^N \mathcal{T}_n(y) + \mathcal{O}_N\left(\frac{1}{y^{N+1}x^{N+1} \log x}\right). \quad (105)$$

Proof sketch. For $Q(z) = q(g(z))$, Theorem 2.2 and $G'(x) = G(x)\psi(x+1)$ give

$$Q(y) = \mathcal{O}(x^{-1}), \quad Q'(y) = \frac{q'(x)}{G'(x)} = \mathcal{O}\left(\frac{1}{yx \log x}\right). \quad (106)$$

The same estimates hold uniformly in a sufficiently small complex neighborhood of y , because g, q , and G are analytic there and the neighborhood in the x -plane shrinks like $1/(yx^2 \log x)$. Thus (104) holds for large y . Repeated use of the operator formula (61), together with fixed-order derivative bounds $q^{(r)}(x) = \mathcal{O}_r(x^{-1})$, gives $\mathcal{T}_n = \mathcal{O}_n(y^{-n}x^{-n}/\log x)$. Taylor's theorem in ε , with the same uniform bounds, yields (105). $\square$

The expansion (105) is in the exponential grade y^{-1} , with exact coefficient functions. If those functions are subsequently replaced by divergent internal expansions, the nested warning of Section 3.3 applies.

7.2 Residual certificates

Formal asymptotics become rigorous numerical approximations once their residual is checked. This is particularly easy here because $\mathfrak{D}$ is uniformly increasing on the principal branch.

Proposition 7.3 (A posteriori error bound). *Let $y \geq 2$, and let $\hat{x} \geq 3$. Define the residual*

$$r = \mathfrak{D}(\hat{x}) - y. \quad (107)$$

Then

$$|\hat{x} - \mathfrak{X}(y)| \leq \frac{|r|}{m_0}, \quad m_0 = \frac{6}{e} \left(\frac{11}{6} - \gamma \right) - \left(\frac{\pi}{4} + \frac{1}{16} \right) > 1.92. \quad (108)$$

If a sharper lower bound $m \leq \mathfrak{D}'(x)$ is known on the interval joining $\hat{x}$ and $\mathfrak{X}(y)$, then m_0 may be replaced by m .

Proof. The proof of [Theorem 2.3](#) gives $\mathfrak{D}'(x) \geq m_0$ for $x \geq 3$. Apply the mean value theorem to $\mathfrak{D}(\hat{x}) - \mathfrak{D}(\mathfrak{X}(y)) = r$. $\square$

For large y , a local derivative estimate is much stronger:

$$\mathfrak{D}'(x) = y \log x (1 + o(1)), \quad x = \mathfrak{X}(y). \quad (109)$$

Thus the error is approximately the residual divided by $y \log x$. One Newton step

$$\hat{x}_{\text{new}} = \hat{x} - \frac{\mathfrak{D}(\hat{x}) - y}{\mathfrak{D}'(\hat{x})} \quad (110)$$

usually converts a short transseries truncation into a high-precision value.

7.3 Balanced truncation

There are two independent truncation indices:

1. M , the number of zero-sector inverse-gamma blocks in [\(47\)](#);
2. N , the number of exponential Lagrange sectors in [\(105\)](#).

The zero-sector error should be made no larger than the last retained exponential sector. If an approximate value $\tilde{g}$ is used in the phase $\cos(\pi g)$, then its absolute error propagates directly into the phase. Consequently, retaining an extremely small y^{-N} correction is pointless unless g has been computed to comparable absolute accuracy. In practice, one either takes enough terms in [\(47\)](#) or performs one Newton correction for $G(x) = y$ before evaluating the exponential sectors.

8 Numerical validation

The following computations used high-precision arithmetic. To isolate the exponential hierarchy, the exact numerical inverse $g(y)$ of G was used inside $\mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3$. The final column separately reports the error of the four-block outer approximation $X - \frac{1}{2} + \sum_{m=1}^4 a_m / X^{2m-1}$.

Table 1: Successive exponential sectors for generic arguments.

y	$ g - \mathfrak{X} $	$ g + \mathcal{T}_1 - \mathfrak{X} $	$ g + \mathcal{T}_1 + \mathcal{T}_2 - \mathfrak{X} $	outer g error
10^2	7.6371×10^{-5}	1.8806×10^{-7}	4.7609×10^{-10}	5.4645×10^{-11}
10^6	3.3545×10^{-8}	4.6652×10^{-18}	2.2653×10^{-22}	2.8457×10^{-13}
10^{20}	1.8356×10^{-23}	7.3241×10^{-45}	2.6668×10^{-66}	2.4032×10^{-16}

The corresponding exact inverse values are

$$\begin{aligned}\mathfrak{X}(10^2) &= 5.4680181917506673490048452082893058\dots, \\ \mathfrak{X}(10^6) &= 9.8768464792043069546910118994035847\dots, \\ \mathfrak{X}(10^{20}) &= 21.542599928440196883351107787710195\dots\end{aligned}$$

Adding $\mathcal{T}_3$ reduces the three errors in the third data column further to approximately 1.23×10^{-12} , 6.08×10^{-30} , and 4.02×10^{-80} , respectively. The table also illustrates balanced truncation: for very large y , a short zero-sector approximation can be the dominant numerical error even after the exponential sectors have become extraordinarily accurate.

At the exact derangement values $y = D_n$, the continuous inverse must return n . The cancellation of the inverse-gamma displacement is shown in Table 2.

Table 2: Recovery of integer arguments at $y = D_n$. Entries are signed errors relative to n .

n	D_n	$g(D_n) - n$	$g + \mathcal{T}_1 - n$	$g + \mathcal{T}_1 + \mathcal{T}_2 - n$
5	44	-1.93565×10^{-3}	3.91873×10^{-6}	2.51266×10^{-8}
10	1334961	2.67167×10^{-8}	9.13999×10^{-16}	-5.19670×10^{-23}
20	895014631192902121	1.68472×10^{-20}	4.43897×10^{-40}	-7.92300×10^{-60}

The leading integer displacement follows directly from the first sector. Because $D_n = G(n) + (-1)^n J(n)$, inversion of G alone gives

$$g(D_n) - n = \frac{(-1)^n J(n)}{G(n)\psi(n+1)} + \mathcal{O}(G(n)^{-2}) \sim \frac{(-1)^n e}{n! \psi(n+1)} \left(\frac{1}{n} - \frac{2}{n^2} + \dots \right). \quad (111)$$

The term $\mathcal{T}_1(D_n)$ cancels this displacement to the next exponential grade.

8.1 A check on optimal Bell truncation

For orientation, truncating (29) near $M \approx x \log x$ gives the errors shown below. The last column divides the error by the scale $\sqrt{x/\log x}/G(x)$ predicted by (34).

Table 3: Late-term scale of the Bell expansion of J .

x	M	Bell truncation error	normalized error
5	8	5.25774×10^{-3}	0.1317
10	23	2.20217×10^{-7}	0.1411
20	60	4.23592×10^{-19}	0.1467
40	148	1.65201×10^{-48}	0.1506

The stable normalized values support the sector-mixing estimate (35). For numerical work, however, the convergent inverse-factorial series (20) removes this issue entirely.

9 Symbolic and numerical construction

The formulas above support several complementary workflows. The choice depends on whether the goal is a formal transseries, a fast numerical value, or a certified integer inverse.

9.1 A robust numerical algorithm

Algorithm 9.1 (Inverse subfactorial for large y). Given $y \geq 2$, an outer order M , and an exponential-sector order N :

1. Compute

$$A = \log \frac{ey}{\sqrt{2\pi}}, \quad w = W_0(A/e), \quad X = A/w,$$

and form the M -block approximation to $g(y)$ from (47).

2. If more accuracy is needed, apply one or two Newton steps to $G(x) - y = 0$:

$$x \leftarrow x - \frac{G(x) - y}{G(x)\psi(x+1)}.$$

The resulting value is the numerical base point $x \approx g(y)$.

3. Evaluate $J(x)$ by the convergent recurrence (23). Evaluate J' , J'' , and higher derivatives either by differentiating the integral or by automatic differentiation of the same convergent series.
4. Form q, q', q'' from (73)–(75), and evaluate $\mathcal{T}_1, \dots, \mathcal{T}_N$ from (61). For $N \leq 3$, use (70)–(72) directly.
5. Return

$$\hat{x} = x + \sum_{n=1}^N \mathcal{T}_n(y).$$

Optionally apply one Newton step (110) to the full interpolation.

6. Compute the residual $r = \mathfrak{D}(\hat{x}) - y$. Use (108), or a sharper local derivative bound, as a certificate.

For the integer generalized inverse $N(y)$, it is enough to certify that the continuous error is smaller than the distance from $\hat{x}$ to the nearest integer boundary. Then $N(y) = \lfloor \hat{x} \rfloor$ by Theorem 2.4.

9.2 Direct symbolic generation of the exponential blocks

The operator formula (61) is ideal for a computer algebra system. The following pseudocode keeps G and q symbolic and generates arbitrary sectors.

```
# Symbolic pseudocode for T_n
# Dx denotes differentiation with respect to x.

def sector(n, G, q, x):
    A = diff(G, x)
    expr = q**n / A
    for k in range(n - 1):
        expr = diff(expr, x) / A
    return (-1)**n * expr / factorial(n)

# Substitute x = g(y) and G(g(y)) = y afterwards.
```

A useful simplification strategy is:

1. replace $G^{(r)}$ by complete Bell polynomials in the derivatives of $\log G$;
2. for the gamma dominant term, replace those derivatives by polygamma functions;
3. factor the resulting power of $G(x) = y$;
4. only then expand J and the polygamma functions asymptotically.

This ordering preserves the exponential grading and avoids premature expansion of the Bell coefficient.

9.3 Symbolic generation of the zero-sector coefficients

The triangular identity (45) is similarly easy to automate. A concise implementation is shown below.

```
# SymPy-style pseudocode for a_m(L)
# kappa(r) is the centered Stirling coefficient.

t, L = symbols('t L')
eta = 0
coefficients = []

for m in range(1, M + 1):
    a = symbol(f'a_{m}')
    trial = eta + a*t**(2*m)
    equation = (trial*(L - 1)
                + (1 + trial)*log(1 + trial))/t
    equation += sum(kappa(r)*t**(2*r - 1)
                    *(1 + trial)**(-(2*r - 1))
                    for r in range(1, m + 1))
    c = coefficient(series(equation, t, 0, 2*m),
                    t, 2*m - 1)
    a_value = solve(c == 0, a)[0]
    coefficients.append(simplify(a_value))
    eta += a_value*t**(2*m)
```

Because the equation at each stage is linear in the new coefficient, this algorithm is stable as a formal calculation and scales substantially better than a blind reversion of the original gamma equation.

9.4 A triangular shift recurrence

There is an alternative to Lagrange–Bürmann inversion. Set

$$x = g(y) + \Delta(y) \quad (112)$$

and expand

$$0 = G(g + \Delta) - G(g) + H(g + \Delta). \quad (113)$$

If $\Delta = \sum_{n \geq 1} y^{-n} U_n(g)$, then comparison of powers of y^{-1} produces a triangular recurrence for U_n . The first equations are

$$0 = \lambda U_1 + H, \quad (114)$$

$$0 = \lambda U_2 + \text{rac12}(\lambda^2 + \lambda') U_1^2 + H' U_1, \quad (115)$$

where all functions are evaluated at $x = g(y)$. Solving gives $U_1 = -H/\lambda$ and the same U_2 as (89). This approach is useful for hand calculations and for proving uniqueness of the coefficient blocks. The operator formula is usually more compact at high order.

10 Examples and extensions of the general theory

10.1 Factorial-type dominant maps

Suppose

$$\log G(x) \sim x \log x - x + \beta \log x + c + \sum_{k \geq 1} \frac{c_k}{x^k}. \quad (116)$$

Then the leading inverse is Lambert- W -controlled. Centering the variable to eliminate as many even or odd correction powers as possible produces a triangular zero-sector recurrence exactly like (45). Once g is known, any additive tail

$$H(x) \sim \sum_{\omega \in \Omega} e^{i\omega x} x^{\rho_\omega} (\log x)^{\tau_\omega} \sum_{k \geq 0} \frac{h_{\omega,k}}{x^k} \quad (117)$$

is inverted by (87). The n -th inverse sector has exponential grade $G(g(y))^{-n} = y^{-n}$, finite frequency support in the n -fold sumset of Ω , and coefficient asymptotics generated by $\lambda^{-1} D_x - k$.

The subfactorial is the model case with $\beta = 1/2$, $\Omega = \{\pi, -\pi\}$, and $H = q \sim x^{-1} \cos(\pi x)$.

10.2 Perturbations of fractional dominant size

Let

$$F(x) = G(x) + G(x)^\alpha h(x), \quad \alpha < 1. \quad (118)$$

The first inverse correction is

$$-\frac{G(x)^\alpha h(x)}{G'(x)} = -G(x)^{-(1-\alpha)} \frac{h(x)}{\lambda(x)}. \quad (119)$$

Thus the natural small parameter is $y^{-(1-\alpha)}$, not y^{-1} . The n -th pure sector has grade $n(1-\alpha)$, while mixed perturbations obey the additive rule (95). Irrational defects produce a noninteger but still well-ordered exponential scale.

10.3 Multiplicative perturbations

A relatively small multiplicative correction

$$F(x) = G(x)(1 + r(x)), \quad r(x) \rightarrow 0, \quad (120)$$

formally corresponds to $\alpha = 1$, so its grade is not supplied by a power of y^{-1} . Instead, the internal scale of r supplies the ordering. The first correction is

$$F^{-1}(y) - g(y) \sim -\frac{r(g(y))}{\lambda(g(y))}. \quad (121)$$

This case is covered by the exact Lagrange formula and by the nested coefficient formalism of (101). It shows why a general transseries theory must track both powers of the dominant function and independent small coefficient scales.

10.4 Several saddles or lateral continuations

Suppose an interpolation has lateral corrections

$$H(x) = \sum_{j=1}^s S_j e^{-\Phi_j(x)} a_j(x), \quad (122)$$

where S_j are Stokes or interpolation parameters. If $e^{-\Phi_j(x)} = G(x)^{-\mu_j} \zeta_j(x)$ with slow ζ_j , then $\alpha_j = 1 - \mu_j$, and Theorem 6.2 gives the complete mixed sector lattice. Products generate actions $\sum_j m_j \Phi_j$, while the block operators generate their prefactors. This is the inversion analogue of the usual additive action semigroup in transseries and resurgence; compare the general framework in [12].

10.5 Interpolation dependence revisited

Let two admissible interpolations be

$$F_1(x) = G(x) + H_1(x), \quad F_2(x) = G(x) + H_2(x), \quad (123)$$

with $H_1(n) = H_2(n)$ at all integers and both $H_j = \mathcal{O}(x^{-1})$. Their inverse zero sectors agree to every algebraic order in X^{-1} , because both are anchored by the same G . Their first difference is

$$F_1^{-1}(y) - F_2^{-1}(y) = -\frac{H_1(g(y)) - H_2(g(y))}{y\lambda(g(y))} + \mathcal{O}(y^{-2}). \quad (124)$$

Thus interpolation ambiguity is itself an exponentially small transseries parameter. The canonical median choice (10) fixes that parameter by a transparent contour prescription.

11 Interpretation of the result

11.1 Why the inverse is much closer to inverse gamma than the function is to gamma

The additive discrepancy $q(x)$ is only algebraically small, $q(x) = \mathcal{O}(x^{-1})$, whereas the derivative of the dominant term is exponentially large:

$$G'(x) = G(x)\psi(x+1) \asymp G(x) \log x. \quad (125)$$

Therefore the inverse displacement is the quotient

$$\mathfrak{X}(y) - g(y) \sim -\frac{q(g(y))}{G'(g(y))} = \mathcal{O}\left(\frac{1}{yg(y) \log g(y)}\right). \quad (126)$$

This elementary derivative mechanism explains the enormous accuracy of inverse gamma as an approximation to inverse subfactorial.

11.2 Why the cosine phase cannot be replaced by $(-1)^x$

At integers, $\cos(\pi n) = (-1)^n$. Away from the integers, however, the real phase is an essential part of the interpolation. The notation $(-1)^x$ is branch-dependent and generally complex. The two lateral continuations naturally carry $e^{\pm i\pi x}$, while their real median carries $\cos(\pi x)$. The finite Fourier structure of all higher sectors is inherited from this choice.

11.3 Universality and nonuniversality

The result separates naturally into two parts.

- *Universal part:* the Lambert-anchored inverse-gamma zero sector, the Lagrange–Bürmann operator calculus, the sector grading rule, and the residual certificates.
- *Interpolation-specific part:* the precise phase and coefficient $q(x) = \cos(\pi x)J(x)$, hence the numerical values of the exponentially small sectors.

For any other interpolation whose additive discrepancy from G is $\mathcal{O}(x^{-1})$, the same zero sector applies and the same operator formulas compute the modified sectors after replacing q .

11.4 What is genuinely transserial here

A finite inverse-gamma expansion alone is an ordinary asymptotic series. The full object contains a strict hierarchy

$$X \gg 1 \gg X^{-1} \gg X^{-2} \gg \dots \gg y^{-1}x^{-1} \gg y^{-2}x^{-2} \gg \dots, \quad (127)$$

with logarithmic coefficients and oscillatory monomials. The hierarchy is closed under multiplication, differentiation, and composition, and its support is locally finite at each exponential grade. These are the defining operational features of a transseries, even though the present positive-real problem requires no Stokes switching once the median interpolation has been fixed.

12 Conclusions

The inverse subfactorial problem has a clean answer only after the continuous object to be inverted is specified. The median incomplete-gamma interpolation

$$\mathfrak{D}(x) = \Gamma(x+1)/e + \cos(\pi x)J(x)$$

is natural, real, exact at every integer, and monotone on the large branch. Its Bell coefficient J admits both a convergent inverse-factorial series and a real-variable Bell expansion with a signed, explicit remainder. This makes it possible to separate numerical evaluation from asymptotic analysis.

The dominant inverse is controlled by a Lambert- W anchor. Centering at $x+1/2$ yields an unusually sparse odd-power expansion, and the coefficient blocks satisfy a simple triangular recurrence. The derangement correction is beyond all orders of that zero sector. Conjugation by the dominant gamma map turns the remaining problem into a near-identity inversion, and the Lagrange-Bürmann formula gives every exponential sector explicitly.

The reusable theory is summarized by two formulas. For a single additive perturbation,

$$U_n = \frac{(-1)^n}{n!} (\lambda^{-1}D_x - (n-1)) \cdots (\lambda^{-1}D_x - 1) \left(\frac{H^n}{\lambda} \right), \quad \lambda = D_x \log G.$$

For several relative scales $G^{\alpha_j} h_j$, a multiindex $\mathbf{m}$ has inverse-sector grade

$$\nu(\mathbf{m}) = \sum_j m_j (1 - \alpha_j).$$

These formulas apply to a broad range of factorial-type, gamma-type, and multi-saddle transseries.

Finally, the Bell coefficient has its own divergent power expansion. Its late terms reach the next exponential grade, so a completely expanded inverse is a nested transseries. Keeping J exact, or evaluating it through its convergent inverse-factorial series, preserves a clean sector decomposition and provides a reliable computational route. This distinction between exact-coefficient sectors and internally expanded sectors is likely the most portable lesson of the subfactorial example.

A Derivation of the centered Stirling coefficients

The shifted Stirling expansion may be written

$$\log \Gamma(u+h) \sim u \log u - u + (h - \tfrac{1}{2}) \log u + \tfrac{1}{2} \log(2\pi) + \sum_{n \geq 1} \frac{(-1)^{n+1} B_{n+1}^{\text{Ber}}(h)}{n(n+1)u^n}, \quad (128)$$

where $B_n^{\text{Ber}}(h)$ is the Bernoulli polynomial. Set $h = 1/2$. Odd Bernoulli polynomials vanish at $1/2$, and

$$B_{2m}^{\text{Ber}}(1/2) = (2^{1-2m} - 1) B_{2m}^{\text{Ber}}. \quad (129)$$

Since $u + h = x + 1$, equations (128)–(129) reduce to (38)–(39).

The first few centered terms are

$$\log \Gamma(x+1) \sim u(\log u - 1) + \frac{1}{2} \log(2\pi) - \frac{1}{24u} + \frac{7}{2880u^3} - \frac{31}{40320u^5} + \frac{127}{215040u^7} - \frac{511}{608256u^9} + \dots \quad (130)$$

B Coefficient extraction for the outer inverse

Let

$$\mathcal{E}(t, \eta) = t^{-1} \{ \eta(L-1) + (1+\eta) \log(1+\eta) \} + \sum_{r \geq 1} \kappa_r t^{2r-1} (1+\eta)^{-(2r-1)}. \quad (131)$$

The coefficient recurrence can be expressed without introducing the temporary symbol ρ_m :

$$a_m(L) = -\frac{1}{L} [t^{2m-1}] \mathcal{E} \left(t, \sum_{j=1}^{m-1} a_j(L) t^{2j} \right). \quad (132)$$

Here $[t^k]$ denotes coefficient extraction. Formula (132) is valid because the omitted contribution from $a_m t^{2m}$ is exactly $La_m t^{2m-1}$.

For example, at first order,

$$0 = La_1 + \kappa_1, \quad a_1 = -\kappa_1/L = 1/(24L).$$

At second order,

$$0 = La_2 + \frac{1}{2} a_1^2 - \kappa_1 a_1 + \kappa_2,$$

which simplifies to (49). Higher orders are no harder conceptually; only the polynomial bookkeeping grows.

C Low-order universal blocks in expanded form

For reference, let $D = D_x$. Expanding (89) gives

$$U_2 = racH H' \lambda^2 - \frac{H^2}{2\lambda} - \frac{H^2 \lambda'}{2\lambda^3}. \quad (133)$$

This agrees with (64) after using $G''/G = \lambda^2 + \lambda'$.

The third block can be left compactly as (90); an expanded form is

$$\begin{aligned} U_3 = & -\frac{H(H')^2}{\lambda^3} - \frac{H^2 H''}{2\lambda^3} + \frac{3H^2 H'}{2\lambda^2} + \frac{3H^2 H' \lambda'}{2\lambda^4} \\ & - \frac{H^3}{3\lambda} - \frac{H^3 \lambda'}{2\lambda^3} - \frac{H^3 (\lambda')^2}{2\lambda^5} + \frac{H^3 \lambda''}{6\lambda^4}. \end{aligned} \quad (134)$$

Substitution $H = q$, $\lambda = \psi(x+1)$ and multiplication by y^{-3} recovers (72) after elementary simplification.

D Bell and Fourier coefficient generation

The power coefficients of J are

$$j_k = (-1)^{k-1} B_k, \quad J(x) \sim \sum_{k \geq 1} j_k x^{-k}. \quad (135)$$

For a lateral mode

$$q_\sigma(x) = \frac{1}{2}e^{i\sigma\pi x}J(x), \quad \sigma \in \{+1, -1\}, \quad (136)$$

one has

$$D_x^r q_\sigma(x) \sim \frac{1}{2}e^{i\sigma\pi x} \sum_{s=0}^r \binom{r}{s} (i\sigma\pi)^{r-s} (-1)^s \sum_{k \geq 1} \frac{\binom{k}{s} j_k}{x^{k+s}}. \quad (137)$$

This formula, together with the Bernoulli expansions of the polygamma functions, gives a direct coefficient-level algorithm for any fixed Fourier mode in any fixed exponential sector.

For the median mode $q = q_+ + q_-$, products are collected by the binomial identity

$$q(x)^n = 2^{-n} J(x)^n \sum_{r=0}^n \binom{n}{r} e^{i(n-2r)\pi x}. \quad (138)$$

This makes the support (81) explicit before any differentiation is performed.

E Notation and hierarchy at a glance

Symbol	Meaning
$D_n = !n$	Number of derangements of n objects.
$G(x)$	Dominant interpolation $\Gamma(x+1)/e$.
$J(x)$	Positive Bell-tail coefficient $\int_0^1 e^{-t}(1-t)^x dt$.
$q(x)$	Oscillatory additive correction $\cos(\pi x)J(x)$.
$\mathfrak{D}(x)$	Canonical real interpolation $G(x) + q(x)$.
$g(y)$	Increasing large inverse of G .
$\mathfrak{X}(y)$	Principal inverse of $\mathfrak{D}$ on $[3, \infty)$.
A, w, X, L	Lambert anchor variables from (43).
p, p_1, p_2	Digamma and first two higher polygamma functions at $x+1$.
λ	Logarithmic derivative G'/G ; for gamma, $\lambda = \psi(x+1)$.
$\mathcal{T}_n$	n -th exponential inverse sector, carrying y^{-n} .
$\mathcal{L}_k$	Universal block operator $\lambda^{-1}D_x - k$.
B_k	Bell number; Bernoulli numbers are written B_k^{Ber} .

The main hierarchy is

$$g(y) = X - \frac{1}{2} + \mathcal{O}(X^{-1}L^{-1}), \quad \mathcal{T}_n(y) = \mathcal{O}_n(y^{-n}g(y)^{-n}(\log g(y))^{-1}). \quad (139)$$

Thus the zero sector is expanded in odd inverse powers of X , while the nonzero sectors are graded by powers of y^{-1} and carry internal power-logarithmic and Fourier expansions.

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